Automated Nutrient and pH Controller for Mycology & Food Security

Title	Automated Nutrient and pH Controller for Mycology & Food Security
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Revision History

Version	Date	Changes
1.0	2025-08-24	Initial Release
1.1	2025-09-02	Integrated optical transmissivity sensor for nutrient concentration monitoring. Updated architecture, hardware, and firmware sections accordingly.

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1.0 Introduction

This document describes the design, implementation, and operation of an automated nutrient and pH controller for hydroponic and liquid culture systems. The system is built on the

Arduino Due platform and is designed for applications in mycology and food security, ensuring optimal environmental conditions for cultivation. This version integrates an optical transmissivity sensor alongside the pH probe to create a dual-sensing system that intelligently doses nutrients based on consumption and maintains precise pH levels. Key features include automated dosing, robust two-point calibration routines for both sensors, and remote control via a Bluetooth interface.

2.0 System Architecture

The system is composed of four main functional blocks:

- **Control Unit:** An **Arduino Due** is the core of the system, selected for its 3.3V logic, high-precision 12-bit Analog-to-Digital Converter (ADC), and extended I/O capabilities.

- **Dosing Subsystem:** Four **12V DC peristaltic pumps** provide precise liquid delivery. Each pump is controlled by a single

  FDS6612A N-Channel MOSFET configured as a low-side switch for efficient power handling.
- **Sensing Subsystem:**
  - A

    pH probe with a 3.3V-compatible amplifier module measures the solution's acidity.
  - An **optical transmissivity (turbidity) sensor** measures the clarity of the solution to infer nutrient concentration.
- **User Interface:**
  - A

    2.2-inch TFT LCD provides a local display for real-time system status.
  - An

    HC-05 Bluetooth module enables remote configuration and monitoring via a mobile device or computer.
  - An

    SD card is used for persistent storage of configuration data.

3.0 Component List (Bill of Materials)

Component	Quantity	Description / Notes
Arduino Due	1	Microcontroller board, 3.3V logic
FDS6612A MOSFET	5	N-Channel, logic-level gate MOSFET for pump control
12V Peristaltic Pump	4	12V DC, ~400mA operating current
1N4001 Diode	4	Flyback diode to protect MOSFETs from inductive spikes
pH Sensor Module	1	3.3V compatible, with BNC connector for probe
Optical Transmissivity Sensor	1	Analog output turbidity sensor, 3.3V compatible.
HC-05 Bluetooth Module	1	Serial (UART) wireless transceiver
2.2" TFT LCD	1	220x176 resolution, with an integrated SD card cage
Power Supply	1	12V DC, minimum 2A to power pumps and board

Component	Quantity	Description / Notes
Breadboard & Jumper Wires	Assorted	For prototyping and connections

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4.0 Hardware Connections

4.1 Peristaltic Pump Control

1. Connect Arduino digital pins

   2, 3, 4, and 5 to the **Gate (G)** pin of each of the four FDS6612A MOSFETs, respectively.
2. Connect the **Source (S)** pin of each MOSFET to the common ground of the circuit. This ground must be shared between the Arduino and the 12V power supply.
3. Connect the

   Drain (D) pin of each MOSFET to the **negative (-)** terminal of its corresponding pump motor.
4. Connect the

   positive (+) terminal of each pump motor to the **+12V** rail of the power supply.
5. Install a **1N4001 diode** in reverse across each pump motor's terminals (cathode to +12V, anode to the MOSFET's Drain). This is critical for preventing back-EMF from damaging the MOSFETs.

4.2 pH Sensing

1. Connect the pH sensor module's

   VCC pin to the Arduino's **3.3V** pin.
2. Connect the module's

   GND pin to the Arduino's **GND** pin.
3. Connect the module's analog output pin (often labeled 'Po' or 'AOUT') to the Arduino's analog input pin

   A0.

4.3 Optical Transmissivity Sensing

1. Connect the optical sensor module's **VCC** pin to the Arduino's **3.3V** pin.
2. Connect the module's **GND** pin to the Arduino's **GND** pin.

3. Connect the module's analog output pin to the Arduino's analog input pin **A1**.

4.4 User Interface

1. **TFT LCD:** Connect the TFT LCD module directly to the Arduino Due's SPI headers. The module's built-in level shifters simplify this connection.
2. **HC-05 Bluetooth Module:**
   - Connect the HC-05

     VCC to the Arduino's **5V** pin.
   - Connect the HC-05

     GND to the Arduino's **GND** pin.
   - Connect the HC-05

     TX pin to the Arduino's **RX1** (Pin 19).
   - Connect the HC-05

     RX pin to the Arduino's **TX1** (Pin 18).
   - **Crucial:** The HC-05's RX pin is not 5V tolerant. You must use a

     voltage divider (e.g., 1kΩ and 2kΩ resistors) or a **bi-directional logic-level shifter** on the line from Arduino TX1 to HC-05 RX to step the 3.3V signal down.

5.0 Firmware & Operation

The system is controlled by open-source firmware written in C++ for the Arduino platform.

5.1 Core Logic

- **Nutrient Dosing (Pumps 1-3):** Dosing is triggered when the optical sensor reading rises above a user-defined transmissivity threshold, indicating that the nutrient solution has become clearer due to consumption.
- **pH Dosing (Pump 4):** Dosing of a pH-Up solution is triggered when the pH sensor reading falls below the user-defined pH target.

5.2 Configuration Management

All system parameters (pH target, transmissivity threshold, pump run times, calibration values) are stored in a `Config` struct in the firmware. These settings can be saved to a text file named `ph_cfg.txt` on the SD card by sending the `SAVE` command via Bluetooth. The system automatically loads this file on startup.

5.3 Calibration Routines

Accurate sensor readings require a two-point calibration for both modules. This is performed by sending commands over the Bluetooth serial interface.

- **pH Calibration:**
  1. Place the pH probe in a pH 7.0 reference solution and send the command:

     `CALPH 7`.
  2. Rinse the probe, place it in a pH 4.0 reference solution, and send:

     `CALPH 4`.
- **Optical Sensor Calibration:**
  1. Place the optical sensor in a fresh, full-strength nutrient solution and send: `CAL_FULL`. This sets the "low transmissivity" baseline.
  2. Clean the sensor, place it in clear water, and send: `CAL_CLEAR`. This sets the "high transmissivity" baseline.

6.0 Maintenance & Troubleshooting

- **Regular Maintenance:** Periodically check that the nutrient and pH solution containers are full. Clean the pH probe tip and the optical sensor lens regularly to prevent buildup and ensure accurate readings.
- **Troubleshooting:**
  - **Pumps Fail to Activate:** Verify the 12V power supply connections and check the MOSFET wiring for shorts or open circuits.
  - **Erratic pH Readings:** The probe may be dirty or the calibration may have drifted. Clean the probe and re-run the two-point calibration.
  - **Erratic Transmissivity Readings:** The sensor lens is likely dirty or has air bubbles on it. Clean the sensor and re-run the `CAL_FULL`/`CAL_CLEAR` calibration.